CYBER.SOHO EDUCATIONAL HUB • BASICS OF COMPUTER NETWORKS

DAY 03

Ethernet & Routing

From the Industry to the Classroom — Building the IT's Future, Today and Together!

Tuesday, May 5, 2026 • 4 contact hours • 2 lectures + 2 labs

AGENDA

Today's Roadmap

L1

LECTURE

How Ethernet Communicates

⏱ 1 hour

E1

EXERCISE

Wireshark Lab

⏱ 1 hour

L2

LECTURE

Routing

⏱ 1 hour

E2

EXERCISE

Project Stage 3 — Wire the LAN

⏱ 1 hour

Reference curriculum: Cisco NetAcad — Networking Essentials 3.0 • Module 21 (Ethernet Switching) + Module 22 (Network Layer)

🎯 **KEY TAKEAWAY** Today is the day Layer 2 and Layer 3 stop being abstract. By 5 pm you will know — beat by beat — what every device does when you press Enter on ping.

LECTURE 1

How Ethernet Communicates

1 hour • MAC addresses, frames, encapsulation, broadcast domains, ARP/ICMP, hierarchy

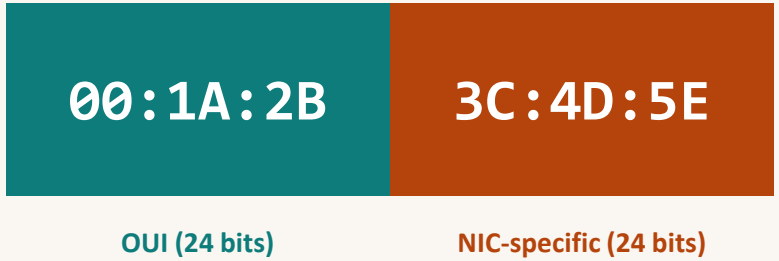
MAC (Media Access Control) Addresses (1/2)

- **48-bit identifier burned into every NIC (Network Interface Card)**
- Format: 00:1A:2B:3C:4D:5E — six pairs of hex
- Operates at Layer 2 of the OSI model

STRUCTURE — split into two halves

- First 24 bits — OUI (Organisationally Unique Identifier)
  - Assigned by the IEEE to a manufacturer
- Last 24 bits — NIC-specific
  - Assigned by the manufacturer to the specific card

ANATOMY



Look up any OUI on the IEEE OUI registry: standards.ieee.org/products-programs/regauth/

MAC (Media Access Control) Addresses (2/2)

TWO IMPORTANT BITS in the FIRST byte of the address

- **Bit 0 — Individual / Group**
  - 0 = unicast (one specific NIC) • 1 = multicast or broadcast
- **Bit 1 — Universal / Local**
  - 0 = factory burn-in (globally unique) • 1 = locally administered (e.g., MAC randomisation)

TRY IT — paste any MAC into 01_mac_address_parser.html

00:50:56:C0:00:01 — VMware vNIC • B8:27:EB:12:34:56 — Raspberry Pi • 02:00:00:00:00:01 — locally administered (U/L bit set)

🎯 **KEY TAKEAWAY** A MAC is a hardware-bound identity. The receptionist delivers to the desk plate — not to the person.

The Ethernet Frame (1/2)

Frame layout — IEEE 802.3 standard

Preamble + SFD	Dest MAC	Source MAC	EtherType	Payload	FCS (CRC-32)
8 B	*6 B*	*6 B*	*2 B*	46 – 1500 B	*4 B*

- Preamble + SFD — receiver clock sync; the NIC strips it (you'll never see it in Wireshark)
- **Destination MAC — read FIRST so the receiver decides in microseconds whether to keep going**
- Source MAC — where to reply to
- EtherType — what's inside: 0x0800 = IPv4 • 0x86DD = IPv6 • 0x0806 = ARP
- Payload — the actual data; 46-byte minimum (padded), 1500-byte maximum (the famous MTU)
- FCS — Frame Check Sequence, a 32-bit CRC; if it fails, the frame is silently dropped

The Ethernet Frame (2/2)

CRITICAL NUMBERS to commit to memory

46	1500	14	4
BYTES	BYTES	BYTES	BYTES
Min payload (padded if smaller)	Max payload (the famous MTU)	Header overhead (after preamble)	FCS / CRC-32 (at the very end)

🎯 **KEY TAKEAWAY** Destination MAC comes FIRST — so the receiver decides in microseconds whether to read the rest. If FCS fails, the frame is silently dropped.

Encapsulation Across the OSI Stack (1/2)

The four-layer journey of a packet — going DOWN the stack

Layer 7	**Application**	*(none — the data)*	`GET /index.html`
Layer 4	**Transport**	*TCP / UDP (ports)*	`TCP src 51234 → dst 80`
Layer 3	**Network**	*IP (IP addresses)*	`src 192.168.1.10 → dst 93.184.216.34`
Layer 2	**Data Link**	*Ethernet (MACs)*	`src AA:BB → dst 00:1A`
Layer 1	**Physical**	*voltage / light / radio*	`bits on the wire`

Encapsulation Across the OSI Stack (2/2)

TWO RULES of encapsulation

1 Going DOWN the stack

Each layer ADDS its own header on the OUTSIDE. Application → Transport → Network → Data Link → Physical.

2 Coming UP the stack

Each layer STRIPS its own header on the WAY IN. Reverses the order — every device handles only what it needs to.

REMEMBER:

Layer 2 changes at every hop. Layer 3 stays the same end-to-end.

🎯 **KEY TAKEAWAY** Driver changes. Address doesn't. Tattoo it — it explains 80% of network behaviour.

Broadcast Domains (1/2)

THE RULE

- **A broadcast frame has destination MAC FF:FF:FF:FF:FF:FF**
- Every NIC on the segment MUST read it — no choice

WHO FORWARDS, WHO BLOCKS?

DEVICE	LAYER	BEHAVIOUR WITH BROADCASTS
Hub	1	Forwards EVERYTHING to every port
Switch	2	FORWARDS broadcasts out every port
Router	3	DROPS broadcasts at the interface
L3 switch	2 + 3	DROPS at routed / VLAN boundaries

Broadcast Domains (2/2)

THE FAMOUS SENTENCE

"A broadcast domain ends at a router (or at a VLAN boundary)."

Every network engineer alive can recite this sentence in their sleep.

THE ANALOGY

Switch = Open Restaurant

Stand up, shout — every diner hears it.
One big broadcast domain.
Every switch port belongs to the same one.

Router = Heavy Door

Shout in one room. The lobby doesn't hear.
To deliver, hand it to the receptionist.
Two separate broadcast domains.

🎯 **KEY TAKEAWAY** Switches = open restaurant. Everyone hears every shout. Routers = heavy door. Shout stays in the room.

ARP, rARP, and ICMP (1/3)

THREE small protocols, THREE small jobs

ARP	**Address Resolution Protocol** • *RFC 826 (1982)* • *Layer 2.5 / 3* *“I have an IP — give me the MAC.”*
rARP	**Reverse ARP** • *RFC 903 (1984)* • *Layer 2.5 / 3* *“I have a MAC — what's my IP? (legacy — replaced by DHCP)”*
ICMP	**Internet Control Message Protocol** • *RFC 792 (1981)* • *Layer 3* *“Network housekeeping — ping, traceroute, error reports.”*

ARP Exchange — Request & Reply (2/3)

THE FLOW — happens millions of times a day on a busy LAN

1. **BROADCAST** — **Host: "Whose IP is X.X.X.X? Tell me."**
   Destination MAC = FF:FF:FF:FF:FF:FF — every NIC reads it

2. **UNICAST** — **Owner of X.X.X.X: "That's me, my MAC is YY:YY:YY."**
   Reply goes only to the asking host — no broadcast needed

3. **CACHE** — **Asking host stores the IP→MAC mapping**
   Cached for ~60 s to several minutes — saves repeat broadcasts

ICMP — The Most-Known Type Pairs (3/3)

TYPES you will meet daily

Type 8 — **Echo Request**
→ *Used by ping*

Type 0 — **Echo Reply**

Type 11 — **Time Exceeded**
→ *Used by traceroute when TTL hits 0*

Type 3 — **Destination Unreachable**
→ *Various sub-codes: network unreachable, host unreachable, port unreachable*

🎯 **KEY TAKEAWAY** ARP is a public shout. Once answered, both sides remember each other for a few minutes. ICMP is the network's housekeeping language.

ARP Tables vs MAC Tables (1/2)

TWO TABLES — easy to confuse, critical to keep straight

ARP TABLE	MAC TABLE (CAM)
End hosts & routers	Switches
MAPS	**MAPS**
IP → MAC	MAC → port
LEARNS BY	**LEARNS BY**
ARP requests / replies	Watching SOURCE MACs of arriving frames
PURPOSE	**PURPOSE**
Build the Layer-2 frame for an outgoing packet	Forward incoming frames out the right port
ANALOGY	**ANALOGY**
Your personal address book	The office switchboard

How a Switch Learns + Aging (2/2)

HOW A SWITCH LEARNS — by watching SOURCE MACs

- Frame arrives on port 3 with source MAC aa:bb:cc:11:22:33
- **Switch records: MAC aa:bb:cc:... lives behind port 3**
- Next frame whose destination matches that MAC → forwarded out port 3 only
- **Unknown destination → flood out every port ("unicast flood")**

AGING — entries don't live forever

- Both tables age out entries — minutes, not hours
- Prevents stale mappings from causing misdelivery

🎯 **KEY TAKEAWAY** ARP table = your address book. MAC table = the switchboard. Different devices, different jobs.

The 3-Layer Hierarchical Design (1/2)

THE THREE TIERS — Cisco's enterprise default

CORE

JOB: Fast backbone between distribution blocks

DEVICE: High-speed L3 switches / routers

→ *Always built REDUNDANT (≥ 2 core switches)*

DISTRIBUTION

JOB: Aggregate access; apply policy (ACL, QoS, inter-VLAN routing)

DEVICE: L3 multilayer switches

→ *Often dual-homed to both core switches*

ACCESS

JOB: Plug in end users (PCs, phones, APs)

DEVICE: Many-port L2 switches

→ *Closest to users — direct user contact*

Why Hierarchy Matters (2/2)

FOUR REASONS to build this way

1 Scalability

Add a new floor → add an access switch and uplink it. No campus redesign.

2 Fault containment

An access failure costs you one classroom — not the whole campus.

3 Predictable performance

Every packet's path is short and known. Latency is bounded.

4 Easier troubleshooting

Failures map cleanly to a tier. "It's a distribution problem."

🎯 **KEY TAKEAWAY** Pulling an access switch costs one classroom. Pulling the core costs the campus. Hence — make the core redundant.

Why We Contain Broadcasts (1/2)

THREE REASONS you actively limit broadcast-domain size

1

PERFORMANCE

Every host inspects every broadcast → CPU interrupts on every machine.
500 chatty hosts × hundreds of broadcasts/s = real load on real CPUs.

2

SECURITY

Broadcasts are visible to EVERYONE on the segment.
ARP and DHCP are classic attack vectors (ARP spoofing, rogue DHCP).
Smaller domain = smaller attack surface.

3

STABILITY

One looping switch / NIC → broadcast storm → segment dies.
Smaller domains contain the damage.

Tools to Contain Broadcasts (2/2)

FOUR TOOLS in your kit

TOOL	WHERE IT CUTS	WHEN TO USE
Router	Between subnets	Always — a router cannot forward a broadcast
VLAN (Virtual LAN)	Inside a switch	Logical separation without buying hardware
Subnetting	At the IP layer	Make smaller IP-level domains
L3 switch	Combo of router + switch	The modern enterprise default

🎯 **KEY TAKEAWAY** Open-plan office is cheap; cubicles cost more but people get work done. Same trade-off, same answer.

EXERCISE 1

Wireshark Lab

1 hour in class + 1 hour at home • Capture frames, decode MAC + EtherType, identify ARP/ICMP, then DHCP

Exercise 1 at a Glance

FORMAT

- 1 hour in class + 1 hour at home
- Optional but HIGHLY recommended
- Counts toward 6-day project

TOOLS

- Wireshark ≥ v4.x
- Your own laptop
- A second device (phone is fine)

LAB ETHICS — HARD RULE

- Capture only on your own machine
- Never the school network or café Wi-Fi
- Sniffing is legal only on networks you control

🎯 **KEY TAKEAWAY** Read the lab in 10 minutes — spend the rest DOING. Reading 2 hours = wasted 2 hours.

Part A — In-Class Steps (60 min)

1. **5 min** — **Pre-flight**
   Pick your active interface; screenshot welcome screen
2. **10 min** — **Capture an Ethernet frame**
   Start capture; run ping -c 4 8.8.8.8; stop
3. **10 min** — **Decode source & destination MAC**
   Note: dest is the GATEWAY's MAC, not Google's. Why? Cross-subnet
4. **10 min** — **Decode EtherType + payload**
   Expand: Ethernet → IPv4 → ICMP, nested
5. **15 min** — **Identify ARP traffic**
   Flush ARP cache; ping gateway; capture request + reply
6. **10 min** — **Identify ICMP traffic**
   Match Echo Requests to Replies by ID + Sequence

🎯 **KEY TAKEAWAY** Destination MAC = the next hop. Never the final destination across a Layer-3 boundary.

Part B — DHCP DORA (1/2)

THE FOUR-MESSAGE DANCE every device runs on join

D

Discover

`src 0.0.0.0 → dst 255.255.255.255`

"Anyone? I have no IP."

O

Offer

`src Server IP → dst 255.255.255.255 / offered`

"Try this IP, lease X, gateway Y, DNS Z."

R

Request

`src 0.0.0.0 → dst 255.255.255.255`

"I'll take THAT server's offer."

A

ACK

`src Server IP → dst Offered IP / broadcast`

"Confirmed. Welcome aboard."

Part B — DHCP DORA (2/2)

TRIGGER DHCP without rebooting

Linux	`sudo dhclient -r && sudo dhclient`
macOS	`Settings → Wi-Fi → TCP/IP → Renew DHCP Lease`
Windows	`ipconfig /release → ipconfig /renew`

WIRESHARK FILTER

`bootp`

(Wireshark still uses the old 'bootp' name — DHCP is built on top of BOOTP)

🎯 **KEY TAKEAWAY** DHCP Discover is broadcast at BOTH Layer 2 and Layer 3 — because the host has no IP yet and doesn't know its server.

Exercise 1 — Deliverable

A 5–10 page report containing

- Cover page — name, date, lab title
- Executive summary (½ page)
- Part A — Ethernet + ARP + ICMP, screenshots + 1 paragraph each step
- Part B — DHCP DORA, the four-message table filled in
- Reflection (½ – 1 page)
- Glossary — every abbreviation expanded

THE LITMUS TEST

"Could a friend who has never seen Wireshark read your report and learn this topic from it?"

If yes — you understand it.

If no — you copied screenshots without thinking.

ALTERNATIVE OPEN-SOURCE TOOLS

tcpdump • tshark • Termshark • Brim/Zui • NetworkMiner

LECTURE 2

Routing

1 hour • Need for routing, routing tables, crossing a Layer-3 boundary

The Need for Routing (1/2)

LAYER 2 ALONE FAILS WHEN…

- **Destination is on a DIFFERENT broadcast domain**
- Switches only know MACs in their own segment
- ARP for a remote IP gets NO answer — no one on this segment owns that IP

ENTER THE ROUTER

- Has interfaces in TWO or more networks at once
- **Looks at destination IP (not MAC)**
- Decides which interface to send out
- **REWRITES the Ethernet header at every hop**
- Leaves the IP header alone (TTL decrements, but IPs persist)

The Need for Routing (2/2) — The Mantra

"IP gets you there.
MAC gets you to the next hop."

Tattoo it. Frame it. Wallpaper your room with it.

It explains 80% of network behaviour.

🎯 **KEY TAKEAWAY** Each router only knows the next hop. None of them knows the whole path. Routing is iterative and local.

Routing Tables — Three Sources (1/3)

EACH ROW: (network, next-hop, exit-interface, metric, source)

C	**CONNECTED** Router has an interface in that subnet — wired into it *AD = 0 • Always wins*
S	**STATIC** A human typed `ip route ...` in running-config *AD = 1 • Survives reboots; doesn't adapt*
Dynamic	**DISCOVERED** RIP / OSPF / EIGRP / BGP discover routes by talking to neighbours *AD = 90 – 200 • Adapts to topology changes*

The Default Route + IOS Codes (2/3)

THE DEFAULT ROUTE — "where everything else goes"

```
S*    0.0.0.0/0 [1/0] via 192.168.12.2
```

- 0.0.0.0/0 = the default route — matches anything not in the table
- S* = the asterisk flags it as 'candidate gateway of last resort'
- [1/0] = [administrative distance / metric]
- via X.X.X.X = the next-hop IP to send unmatched traffic to

IOS CODES YOU WILL MEET DAILY

- C = connected • L = local • S = static • S* = static default
- R = RIP • O = OSPF • D = EIGRP • B = BGP

🎯 **KEY TAKEAWAY** When two sources offer the same route, the one with the lower administrative distance wins.

Administrative Distance Cheat Sheet (3/3)

STANDARD CISCO DEFAULTS — lower wins

SOURCE	AD
Directly connected	0
Static	1
eBGP (external BGP)	20
EIGRP (internal)	90
OSPF	110
RIP	120
iBGP (internal BGP)	200
Unknown / unreachable	255

🎯 **KEY TAKEAWAY** Connected (0) beats Static (1) beats every dynamic protocol. Static (1) beats OSPF (110) — even when OSPF has fresher data.

Crossing a Layer-3 Boundary (1/2)

WHAT STAYS THE SAME end-to-end	WHAT CHANGES at every hop
▸ Destination IP — the goal ▸ Source IP (unless NAT — Day 4) ▸ TCP / UDP payload	▸ Source MAC — this hop's outbound port ▸ Destination MAC — next-hop's MAC, learned via ARP ▸ TTL — decrements by 1 ▸ Ethernet FCS — recomputed (header changed)

MENTAL MODEL: the package's address label = IP destination (never altered).

The truck driver and license plate = MAC addresses (changed at every border).

End-to-End Trace — IP Constant, MAC Mutates (2/2)

FOLLOW the frame from Host A through three routers to Host B

WHAT EACH HOP SEES

src IP	10.0.0.50	10.0.0.50	10.0.0.50	10.0.0.50
dst IP	20.0.0.50	20.0.0.50	20.0.0.50	20.0.0.50
src MAC	Host A	R1 wan	R2 wan	R3 lan
dst MAC	R1 lan	R2 wan	R3 wan	Host B
TTL	64	63	62	61

🎯 **KEY TAKEAWAY** Driver changes. Address doesn't. That is the whole story of inter-LAN communication.

EXERCISE 2

Project Stage 3

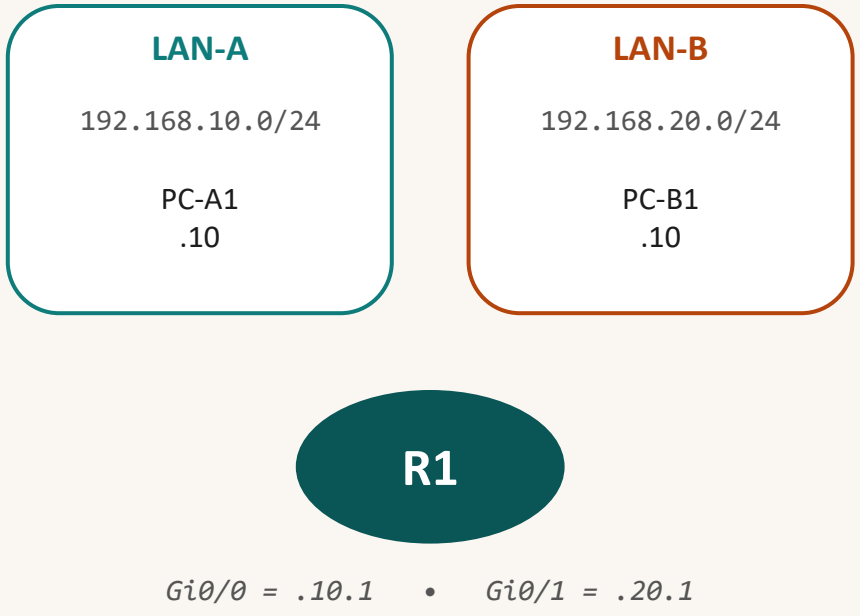
1 hour in class + 1 hour at home • Add a second LAN segment, configure router interfaces, verify ping

Exercise 2 at a Glance

GOAL

- Take Day-2 GNS3 topology
- Add a second LAN segment (Switch-B + PC-B1)
- Configure router interfaces on both segments
- **Verify ping between segments**

TOPOLOGY



TOOLS

- GNS3 ≥ v2.2
- Cisco IOS image (or VyOS / FRR alternative)
- Your Day-2 GNS3 project

🎯 **KEY TAKEAWAY** Stage 3 of the running 6-day project. Don't skip it — Day 4 starts from this exact state.

Part A — In-Class Steps (60 min)

1	5 min	Open Day-2 GNS3 project
2	10 min	Add Switch-B + PC-B1 to the canvas
3	15 min	Configure R1's two interfaces (Gi0/0 = .10.1, Gi0/1 = .20.1, no shutdown)
4	10 min	Configure both PCs (IP, mask, gateway via 'ip ... save')
5	5 min	Verify directly-connected routes — show ip route → 2× C, 2× L
6	10 min	Verify ping between segments — ping 192.168.20.10 from PC-A1
7	5 min	Capture in Wireshark — see gateway MAC as L2 dest, PC-B1 IP as L3 dest

🎯 **KEY TAKEAWAY** Step 7 is the SINGLE MOST IMPORTANT screenshot of the day. It proves you understand 'IP gets you there, MAC gets you to the next hop.'

Part B — At-Home Steps (60 min)

8	15 min	Add a third host PC-A2 on LAN-A — verify same-segment ping (no router) and cross-segment ping (with router)
9	10 min	Inspect ARP cache on PC-A1 vs R1 — PC-A1 sees only the gateway; R1 sees hosts on both segments
10	15 min	Document topology + IP plan + change log
11	10 min	Stretch goal — add a static route, observe Destination Host vs Network Unreachable
12	10 min	Save & export GNS3 project — Day 4 starts from here

🎯 **KEY TAKEAWAY** PC-A1's ARP cache stays small because it has only ever talked to its gateway — never to a remote host directly.

WRAP-UP

Day 03 Mental Model — One Diagram

Same subnet?

✅ YES — same subnet

- ARP for the destination directly
- Build frame to that MAC
- Switch delivers it locally
- No router involvement

🔁 NO — different subnet

- ARP for the GATEWAY
- Frame to gateway's MAC, IP unchanged
- Router decides next hop
- L2 header rewritten at every hop

🎯 **KEY TAKEAWAY** If you can draw this slide from memory tomorrow, you've got Day 03 nailed.

WRAP-UP

Three Sentences to Remember Forever

1. ***A switch forwards broadcasts; a router blocks them.***

2. ***IP gets you there; MAC gets you to the next hop.***

3. ***Every router only knows the next hop, never the whole path.***

LOOKING AHEAD

What's Next — Day 4

Thursday May 7, 2026 • IP Addressing & Address Management

TOPICS

- IPv4 — 32-bit, dotted-decimal, binary ↔ decimal
- Subnet masks, classes (A–E), public vs private (RFC 1918)
- Unicast / multicast / broadcast
- DHCP — DORA dance (revisited from today)
- NAT (Network Address Translation), PAT (Port Address Translation)
- Why IPv6 will replace IPv4 — address exhaustion, simpler header

READING BEFORE DAY 4

- Cisco NetAcad — NetEss 3.0
  - Module 21 — Ethernet Switching
  - Module 22 — Network Layer
- Repeat your Wireshark capture against DHCP
- **Keep your GNS3 project saved!**

🎯 **KEY TAKEAWAY** Day 03's project state is the starting point of Day 04. Save it now. Export it now.

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LAB ETHICS

All techniques used only on your own GNS3 topology. Verify any installer at virustotal.com before running. Cyber.SoHo Educational Hub assumes no liability for misuse.

Thank You

Day 03 — Ethernet & Routing • Complete

Ask-anything time. Then save your GNS3 project.

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